Q5

# Photochemistry – Chapter-3

(2022)

3Q What do you mean by sensitized fluorescene and quenching of fluorescence? (1.5)

Ans:

## Sensitized fluorescence:

A substance which is normally non-fluorescent may be made fluorescent in the presence of other fluorescent substances. This phenomenon is known as sensitized fluorescence.

For example if the vapour of Tl is added to Hg vapour and then exposed to the radiations of wavelength 2537Å, the vapour of Tl fluorescence

$$Hg + h\nu \rightarrow Hg^*$$

$$Hg^* + Tl \rightarrow Tl^* + Hg$$

$$Tl^* \longrightarrow Tl + h\nu$$

Scanned with ACE Scanner

Quenching of Fluorescence: When a photochemically excited atom has a chance to undergo collison with another atom or a molecule before it fluoresces, the intensity of the fluorescent radiation may be diminished or stopped. This phenomenon is known as quenching of fluorescence.

Example: The excited $Hg^*$ atom undergo collison with $H_2$ molecule and thus it dissociates.

$$Hg^* + H_2 \longrightarrow Hg + 2H$$

(b) Define photosensitized reactions and give examples using different types of photosensitizer. (3) (2017) (2020)

Answer

Photosensitization: Certain reaction are known which are not sensitive to light.

Scanned with ACE Scanner

These reactions can be made light sensitive by adding a small amount of foreign material which can absorb light and stimulate the reaction without itself taking part in the reaction. Such an added material is known as photosensitizer and the phenomenon as photosensitization.

Examples

① Reactions sensitized by Mercury atoms:-

Dissociation of $H_2$ into atom at 253.7 nm in the presence of Hg vapour was studied by Cario and Frank (1922).

$$Hg(^1S_0) + H_2 \longrightarrow 2H + Hg(^1S_0) \text{ at } 253.7\text{ nm}$$

② Chlorine as a photosensitizer:-

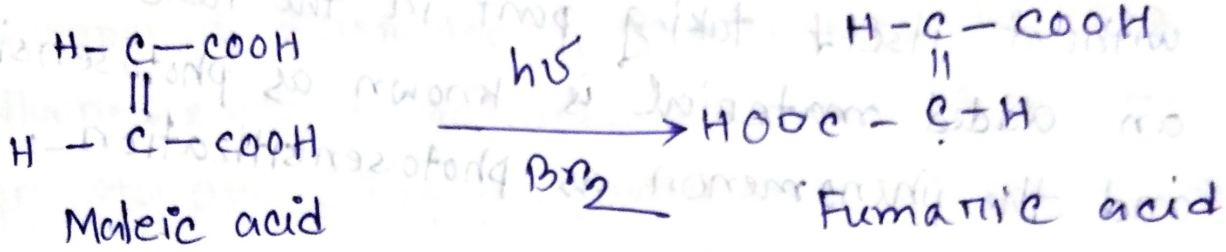
Decomposition of ozone in presence of chlorine in UV.

$$2O_3 + h\nu \xrightarrow{Cl_2} 3O_2$$

Scanned with ACE Scanner

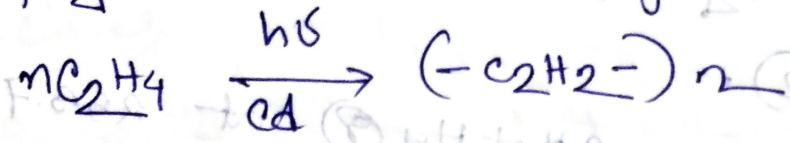
③ Bromine as a photosensitizer:-

Bromine acts as a photosensitizer in the conversion of maleic acid into fumaric acid.



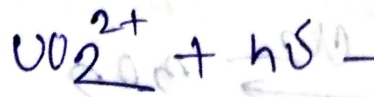

④ Cadmium vapour as a photosensitizer:-

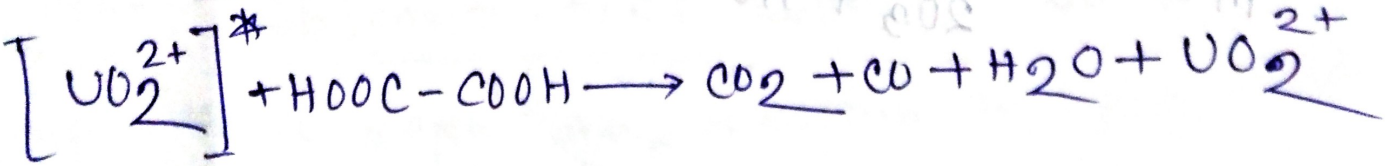
Cadmium vapour acts as a photosensitizer for the polymerisation of ethylene.




⑤ Uranyl ion as a photosensitizer:-

Uranyl ion acts as a photosensitizer in the photolysis of formic acid.




Scanned with ACE Scanner

(b) Chlorophyll as a photosensitizer:-

Chlorophyll acts as a photosensitizer in the photosynthesis of carbohydrate from $CO_2$ and $H_2O$.

$$\text{Chlorophyll} + h\upsilon \longrightarrow [\text{Chlorophyll}]^*$$

$$6CO_2 + H_2O + [\text{Chlorophyll}]^* \rightarrow C_6H_{12}O_6 + O_2 + \text{Chlorophyll}$$

(c) Discuss the primary photochemical process using molecular spectra: (3.5) (2020)

Answer

Primary photochemical process:- Process in which light radiation is absorbed by an atom or molecule giving rise to the formation of an excited atom or an excited molecule.

$$A + h\upsilon \longrightarrow A^*$$

First of all photochemical reactions requires absorption of energy. The reacting molecule is raised to a higher energy level. The nature depends on the relationship between the upper and lower electronic states of the molecule.

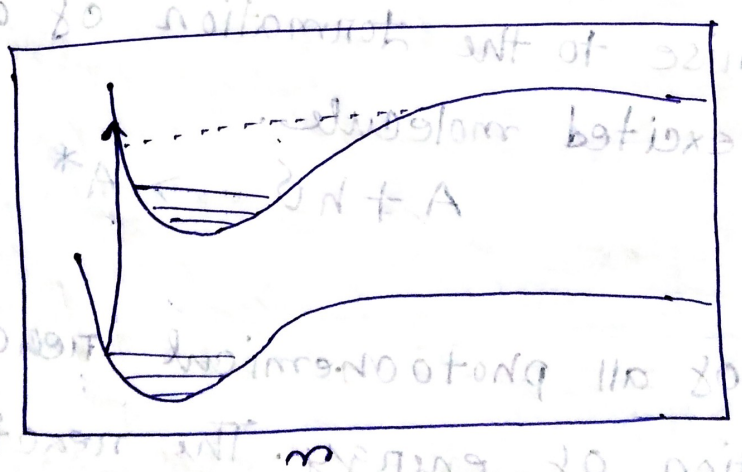
Scanned with ACE Scanner

And the reactions occurs in the visible or ultraviolet region.

Four types of behaviour are of interest in connection with the primary photochemical process :—

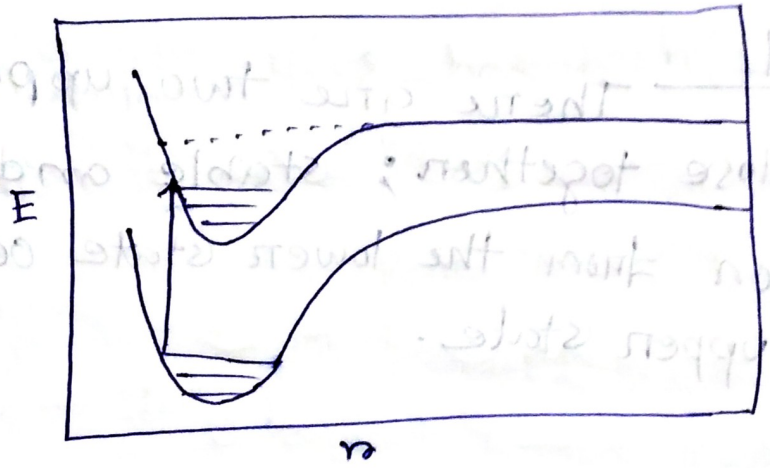
Type-1 The nature of the potential energy curve is such that in the transition indicated the vibrational energy of the molecule in the upper electronic state exceeds the maximum value.

After the absorption of energy the molecule will dissociate in its 1st oscillation.



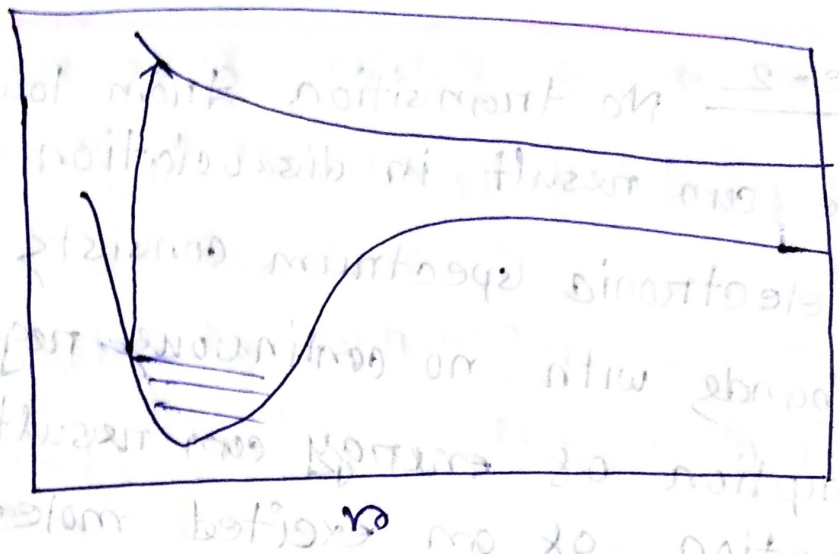

Scanned with ACE Scanner

**Type-2** No transition from lower to upper state can result in dissociation of the molecule. The electronic spectrum consists of a series of bands with no continuous region. Absorption of energy can result only the formation of an excited molecule.

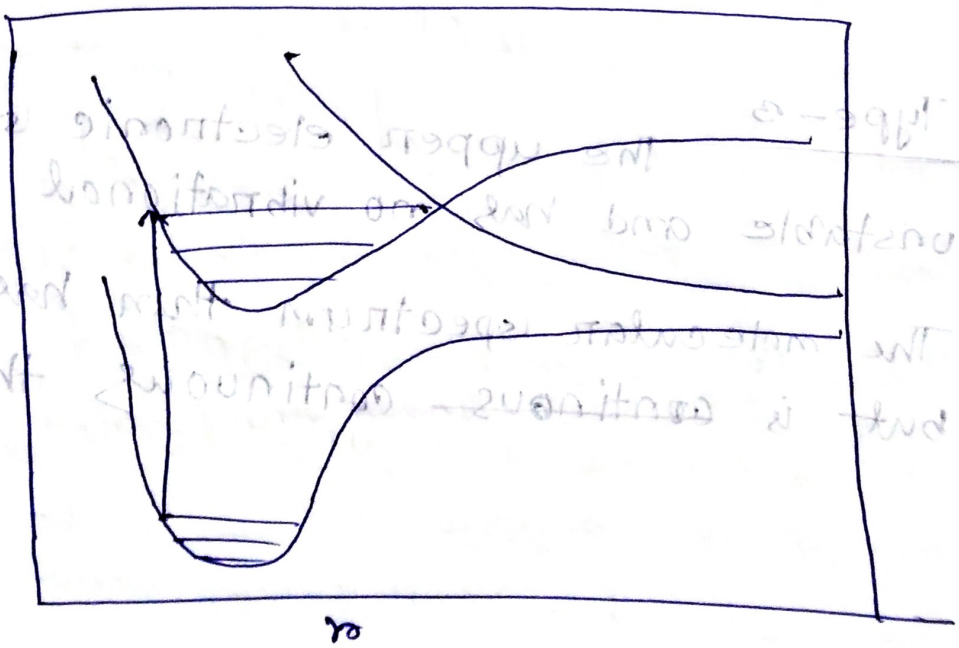



**Type-3** The upper electronic state is completely unstable and has no vibrational levels. The molecular spectrum then has no bands, but is ~~continuous~~ continuous throughout.

Scanned with ACE Scanner




Type-4 — There are two upper electronic levels close together; stable and unstable. Transition from the lower state occurs to the stable upper state.




Scanned with ACE Scanner

(d) Deduce an expression for the kinetics of photochemical reaction between $H_2$ and $Br_2$.

2020-4 2017-4 3.5

Answer:

The overall photochemical reaction between $H_2$ and $Br_2$ is

$$H_2 + Br_2 \xrightarrow{h\nu} 2HBr$$

(i) The mechanism that has been proposed for this reaction is as follows.

| | Rate constant | Rate law |
|---|---|---|
| 1. Initiation | | |
| (i) $Br_2 + h\nu \rightarrow 2Br$ | $k_1$ | Rate $= k_1 I_{abs}$ |
| 2. Propagation | | |
| (ii) $Br + H_2 \rightarrow HBr + H$ | $k_2$ | Rate $= k_2[Br][H_2]$ |
| (iii) $H + Br_2 \rightarrow Br + H$ | $k_3$ | Rate $= k_3[H][Br_2]$ |
| 3. Inhibition | | |
| (iv) $H + HBr \rightarrow Br + H_2$ | $k_4$ | Rate $= k_4[H][HBr]$ |
| 4. Termination | | |
| (v) $Br + Br \rightarrow Br_2$ | $k_5$ | Rate $= k_5[Br]^2$ |

Scanned with ACE Scanner

(iii) Deduction of Rate Law

$$\frac{d[HBr]}{dt} = k_2[H_2][Br] + k_3[H][Br_2] - k_4[H][HBr] \quad \text{——(1)}$$

under steady-state condition

$$\frac{d[H]}{dt} = k_2[Br][H_2] - k_3[H][Br_2] - k_4[H][HBr] = 0 \quad \text{——(2)}$$

and $$\frac{d[Br]}{dt} = k_1 I_{abs} - k_2[Br][H_2] + k_3[H][Br_2] + k_4[H][HBr] - k_5[Br]^2 = 0 \quad \text{——(3)}$$

Adding equations (2) and (3), we get

$$k_1 I_{abs} - k_5[Br]^2 = 0$$

$$\therefore [Br] = \left[\frac{k_1}{k_5} \cdot I_{abs}\right]^{1/2}$$

From equation (2), we get

$$[H] = \frac{k_2[Br][H_2]}{k_3[Br_2] + k_4[HBr]} \quad \text{——(5)}$$

Scanned with ACE Scanner

Putting value of $[Br]$ from (4) in (5), we get

$$[H] = \frac{K_2\left(\frac{K_1}{K_5} I_{abs}\right)[H_2]}{K_3[Br_2] + K_4[HBr]}$$

From equations (1), (4) and (6), we get on simplification

$$\frac{d[HBr]}{dt} = \frac{2K_2\sqrt{K_1 \cdot I_{abs}/K_5}\cdot[H_2]}{I + (K_4[HBr]/K_3[Br_2])} \quad \text{—— (7)}$$

This is the rate-law equation for over-all reaction.

(e) The dissociation energy of a molecule is 142.95 kcal $mol^{-1}$. What is the longest wavelength of light capable of dissociation of this molecule? (2.5)

Answer We know

$$E = \frac{Nhc}{\lambda}$$

$$\Rightarrow \lambda = \frac{Nhc}{E}$$

where,

$E = 142.95$ kcal/mol

$= 142.95 \times 10^3 \times 4.184$ J/mol

$N = 6.023 \times 10^{23}$

$c = 3 \times 10^8$ $ms^{-1}$

$h = 6.626 \times 10^{-34}$ J/s

$\lambda = ?$

Scanned with ACE Scanner

$$= \frac{6.023 \times 10^{23} \times 6.626 \times 10^{-34} \times 3 \times 10^{8}}{142.95 \times 10^{3} \times 4.184}$$

$$= 2001.75 \times 10^{-10} \text{ m}$$

$$= 2001.75 \text{ A}^{\circ}$$

2024

3@ State the Einstein's law of photochemistry. (1)

2018 2017

Answer:

Einstein law of photochemical equivalence:-

The law states that

"When an atom or molecule absorbs light of a given frequency, it absorbs one quantum only."

This law also states that (stark and Einstein)

"Each molecule which takes part in a chemical reaction absorb one quantum of light which induces the reaction.

Scanned with ACE Scanner

Explanation

A molecule acquire energy by absorbing photon as

$$A + h\upsilon = A^*$$

The energy of one photon is equal to $h\upsilon$, where $\upsilon$ is the frequency of absorbing photon and h is the plank's constant.

(b) What do you mean by quantum yield? which factors are responsible for low and high quantum yield? (3) (2018)

Answer:

Efficiency of a photochemical process is expressed in terms of quantum efficiency or quantum yield ($\Phi$). It states that

"The number of molecules reacting per quantum of light absorbed.

$$\Phi = \frac{\text{Number of molecules reacting in a given time}}{\text{Number of quanta of radiation absorbed in the same time.}}$$

Scanned with ACE Scanner

Energy of one photon $E = h\nu$, and 1 mole = N molecules. Since each molecule absorbs one photon.

* Reasons for low quantum yield:-

⇒ Excited molecules may get deactivated before they form a product so the value is less than 1.

⇒ Collision of excited molecule with non-excited molecule.

⇒ Molecules may not receive adequate energy to enable them to react.

⇒ The photochemical reaction may be reserved.

* Reasons for high quantum yield:-

⇒ One photon absorbed in a primary reaction dissociates one molecule of the reactants. But the excited atoms that result may start a subsequent secondary reaction in which a

Scanned with ACE Scanner

further molecule is decomposed

$AB + h\nu \rightarrow A + B$ (primary)

$AB + A \rightarrow A_2 + B$ (secondary)

⇒ When there are two or more reactants, a molecule of one of them absorbs a photon and dissociates (primary reaction). The excited atom that is produced starts a secondary reaction chain

$A_2 + h\nu \rightarrow 2A$ (primary)

$A + B_2 \rightarrow AB + B$ (secondary)

$B + A_2 \rightarrow AB + A$ (Reaction chain)

⇒ The radiation absorption includes the production of the atoms with initiate the chain reaction.

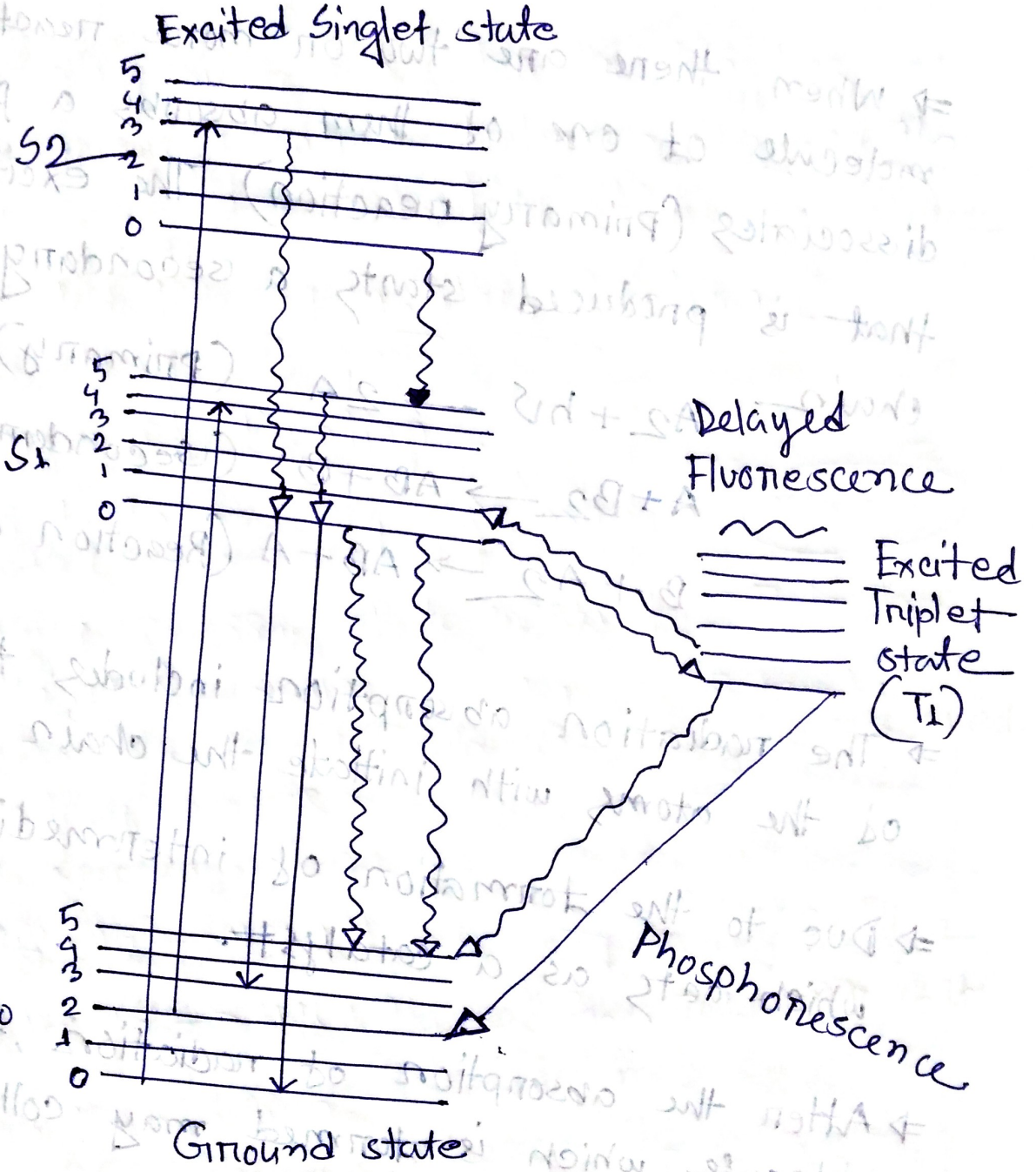
⇒ Due to the formation of intermediate product which acts as a catalyst.

⇒ After the absorption of radiation, the activated molecule which is formed may collide with another active molecule.

Scanned with ACE Scanner

© Explain the terms fluorescence and phosphorescence using Jablonski diagram. (3)

Answer: Jablonski Energy Diagram




Scanned with ACE Scanner

## Fluorescence :-

Fluorescence is the emission of light by a substance that has absorbed electromagnetic radiation. During fluorescence light is absorbed at a certain wavelength and should be emitted at a greater wavelength.

For example

Chlorophyll present in green leaves shows the phenomenon of fluorescence.

Petroleum, vapours of iodine, acetone and hydrocarbons (Paraffin and olefins) have been found to fluoresce in ultraviolet regions.

## Phosphorescence :-

When a beam of light is incident on certain substances, they emit light continously even after the incident light is cut off. This type of delayed fluorescence is called phosphorescence.

Scanned with ACE Scanner

For example
Sulphates of calcium, barium, and strontium exhibit phosphorescence.

(d) Deduce an expression for the kinetics of photochemical reaction between $H_2$ and $Cl_2$. (4·5)

Answer:

## Kinetics of $H_2 + Cl_2$ Reaction

Mechanism of Reaction

Case-1. In absence of oxygen (2018-3)

| | | Rate constant | Rate law |
|---|---|---|---|
| (a) Primary process | | | |
| (i) | $Cl_2 + h\nu \longrightarrow 2HCl$ | $k_1$ | Rate $= k_1 I_{abs}$ |
| (b) Secondary process | | | |
| 1. Chain propagation | | | |
| (ii) | $Cl + H_2 \longrightarrow HCl + H$ | $k_2$ | Rate $= k_2 [Cl][H_2]$ |
| (iii) | $H + Cl_2 \longrightarrow HCl + Cl$ | $k_3$ | Rate $= k_3 [H][Cl_2]$ |
| 2. Chain termination | | | |
| (iv) | $Cl +$ wall $\longrightarrow \frac{1}{2} Cl_2$ | $k_4$ | Rate $= k_4 [Cl]$ |

Scanned with ACE Scanner

The above mechanism is interpreted with the help of steady-state principle.

∴ Net rate of formation of Cl = (Rate formation of Cl − Rate of removal of Cl)

$$\frac{d[Cl]}{dt} = K_1 I_{abs} + K_3 [H][Cl_2] - K_2 [Cl][H_2] - K_4 [Cl]$$

But at steady-state: $\frac{d[Cl]}{dt} = 0$

$$\therefore K_1 I_{abs} + K_3 [H][Cl] - K_2 [Cl][H_2] - K_4 [Cl] = 0 \quad \text{—} (1)$$

Similarly, $\frac{d[H]}{dt} = K_2 [Cl][H_2] - K_3 [H][Cl] = 0 \quad \text{—} (2)$

Adding equations (1) and (2), we get

$$K_1 I_{abs} - K_4 [Cl] = 0$$

$$[Cl] = \frac{K_1}{K_4} \cdot I_{abs} \quad \text{—} (3)$$

From equation (2), $K_2 [Cl][H_2] = K_3 [H][Cl_2] \quad \text{—} (4)$

Scanned with ACE Scanner

Net rate of formation of HCl is given by

$$\frac{d[HCl]}{dt} = k_2[Cl][H_2] + k_3[H][Cl_2] \quad \text{——— (5)}$$

From equations (4) and (5)

$$\frac{d[HCl]}{dt} = 2k_2[Cl][H_2] \quad \text{——— (6)}$$

From equations (3) and (6), we get

$$\frac{d[HCl]}{dt} = 2\frac{k_1 \cdot k_2}{k_4} \cdot I_{abs}[H_2] \quad \text{——— (7)}$$

Equation (7) is the rate-law of the over-all reaction. This equation is in full agreement with experimental results.

**Case-2** In presence of oxygen

| | Rate constant | Rate law |
|---|---|---|
| @ Primary Process: | | |
| (i) $Cl_2 + h\nu \longrightarrow 2HCl$ | $k_1$ | $Rate = k_1 \cdot I_{abs}$ |

Scanned with ACE Scanner

(b) Secondary process

(ii) $Cl + H_2 \longrightarrow HCl + H$ — $k_2$ — Rate $= k_2[Cl][H_2]$

(iii) $H + Cl_2 \longrightarrow HCl + Cl$ — $k_3$ — Rate $= k_3[H][Cl_2]$

(iv) $H + O_2 \longrightarrow HO_2$ — $k_4$ — Rate $= k_4[H][O_2]$

(v) $Cl + O_2 \longrightarrow ClO_2$ — $k_5$ — Rate $= k_5[Cl][O_2]$

(vi) $Cl + X \longrightarrow ClX$ — $k_6$ — Rate $= k_6[Cl][X]$

In step (vi) X is any substance which removes chlorine atoms. Applying steady-state principle to Cl-atoms

$$\frac{d[Cl]}{dt} = k_1 \cdot I_{abs} - k_2[Cl][H_2] + k_3[H][Cl_2] - k_5[Cl][O_2] - k_6[Cl][X] = 0$$

$$\therefore [Cl] = \frac{k_1 \cdot I_{abs} + k_3[H][Cl_2]}{k_2[H_2] + k_5[O_2] + k_6[X]} \quad \text{— (1)}$$

Applying steady-state principle to H-atoms

$$\frac{d[H]}{dt} = k_2[Cl][H_2] - k_3[H][Cl_2] - k_4[H][O_2] = 0$$

Scanned with ACE Scanner

$$\therefore [Cl] = \frac{k_3[H][Cl_2] + k_4[H][O_2]}{k_2[H_2]} \quad \text{---(2)}$$

Equating equations (1) and (2)

$$\frac{k_1 \cdot I_{abs} + k_3[H][Cl_2]}{k_2[H_2] + k_5[O_2] + k_6[X]} = \frac{k_3[H][Cl_2] + k_4[H][O_2]}{k_2[H_2]}$$

Neglecting $k_4 \cdot k_5 \cdot [H][O_2]$ term, which is indeed very small, the above equation is simplified to give value of $[H]$. Thus

$$[H] = \frac{k_1 \cdot I_{abs}\, k_2[H_2]}{k_3 k_6[Cl_2][X] + [O_2]\{k_2 k_4[H] + k_3 k_5[Cl_2] + k_4 k_6[X]\}} \quad \text{---(3)}$$

For the rate of formation of HCl, we have to consider the steps (ii) and (iii). Since step (ii) is very-very slow, it has no significant contribution towards formation of HCl. Hence step (iii) will mainly provide HCl.

Scanned with ACE Scanner

Therefore $$\frac{d[HCl]}{dt} = k_3 [H][Cl] \quad \text{——(4)}$$

Putting values of [H] from equation (3) in (4),

$$\frac{d[HCl]}{dt} = \frac{k_1 k_2 k_3 I_{abs} [H_2][Cl_2]}{k_3 k_6 [Cl_2][X] + [O_2]\{k_2 k_4 [H_2] + k_3 k_5 [Cl_2] + k_4 k_6 [X]\}}$$

Dividing numerator and denominator by $k_2 \cdot k_4$ we get

$$\frac{d[HCl]}{dt} = \frac{\left(\frac{k_1 k_3}{k_4}\right) I_{abs} [H_2][Cl_2]}{\left(\frac{k_3 k_6}{k_2 k_4}\right)[Cl_2][X] + [O_2]\left([H_2] + \frac{k_3 k_5}{k_2 k_4}[Cl_2] + \frac{k_6}{k_2}[X]\right)} \quad \text{——(5)}$$

This is equation of the rate-law of over-all reaction and is in full agreement with experimental results.

Scanned with ACE Scanner

(e) The dissociation energy of hydrogen is 102.9 Kcal/mol. If $H_2$ is dissociated by illumination with radiation of wavelength 2537 A°, what fraction of radiant energy will be converted into K.E? (2.5)

**Answer:**

Here,

Dissociation energy per mole $= \frac{102.9 \times 10^3 \times 4.184}{6.023 \times 10^{23}}$

$= 7.15 \times 10^{-19}$ J

Again,

Energy of one quantum $= \frac{hc}{\lambda}$

$= \frac{6.626 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{2537 \times 10^{-10} \text{ m}}$

$= 7.84 \times 10^{-19}$ J

∴ Energy converted into K.E $= (7.84 \times 10^{-19} - 7.15 \times 10^{-19})$ J

$= 0.69 \times 10^{-19}$ J

Scanned with ACE Scanner

∴ Fraction converted into K.E = $\frac{0.69 \times 10^{-19}}{7.84 \times 10^{-19}}$

= 0.088

2020

3@ Differentiate between photochemical reaction and thermal reaction ? (1.5)

Answer

| Photochemical reaction | Thermal Reaction |
|---|---|
| 1. These reactions involve absorption of light radiations. | 1. These reactions involve absorption or evolution of heat. |
| 2. Presence of light is the primary requirement. | 2. These reactions can take place in light as well as in dark. |
| 3. Rate of reaction is independent on temperature. | 3. Rate of reaction depends on temperature. |
| 4. $\Delta G$ may be + (ve) or − (ve) | 4. $\Delta G$ is always − (ve). |
| 5. Photochemical activation is highly selective process | 5. Thermal activation is not selective in nature. |
| 6. Energy varies from 23 to 230 kcal/mol. | 6. Energy needed in the range of 100 to 1000 kcal/mol. |

Scanned with ACE Scanner

Q. The dissociation energy of a molecule is 142.95 kcal/mol. What is the longest wavelength of light capable of dissociation of this molecule. (2)

Answer

Here

Dissociation energy per molecule $= \frac{142.95 \times 10^3 \times 4.184}{6.023 \times 10^{23}}$

$= 9.93 \times 10^{-19}$ J

Again energy of one quantum $E = \frac{hc}{\lambda}$

$\Rightarrow \lambda = \frac{hc}{E}$

$= \frac{6.626 \times 10^{-34} \text{ Js} \times 3 \times 10^8 \text{ ms}^{-1}}{9.93 \times 10^{-19} \text{ J}}$

$= 2 \times 10^{-7}$ m

2018

3@ State the Grotthus-Draper law of photochemistry. (1)

2017

Answer

Grotthuss-Draper law states that

"Only those radiation which are absorbed by a reacting substance or system are responsible for producing chemical change".

According to this law, all light radiations are not bringing the chemical reaction. Some are ~~bringing~~ increase the kinetic energy of molecule while some are re-emitted (fluorescence).

(c) Why does the value of quantum yield deviate from unity? Give examples where the quantum yield is greater than unity and less than unity. (3)

Answer

Einstein is the unit of photochemical intensity. The value of $\phi$ may vary from 1 to $10^{10}$; in some cases it is less than 1.

cases

(1) For a reaction that obeys strictly the Einstein law, one molecule decomposes per photon, the

Scanned with ACE Scanner

quantum yield, $\phi = 1$.

(ii) When two or more molecules are decomposed per photon, $\phi > 1$ and the reaction has a high quantum yield.

⇒ There are two reasons for the deviation of quantum yield from unity.

⇒ Examples of high quantum yield reactions are

i) The combination of CO and $Cl_2$ in the presence of light

$$CO + Cl_2 \xrightarrow{h\nu} COCl_2, \quad \phi = 10^3$$

ii) The combination of hydrogen ~~chloride~~ and $Cl_2$ to form HCl by a wavelength less than 4500 A°

$$H_2 + Cl_2 \xrightarrow{h\nu} 2HCl, \quad \phi = 10^4 - 10^6$$

⇒ Examples of low quantum yield reactions are

i) The combination of $H_2$ and $Br_2$ to form HBr.

$$H_2 + Br_2 \xrightarrow{h\nu} 2HBr, \quad \phi = 0.01$$

Scanned with ACE Scanner

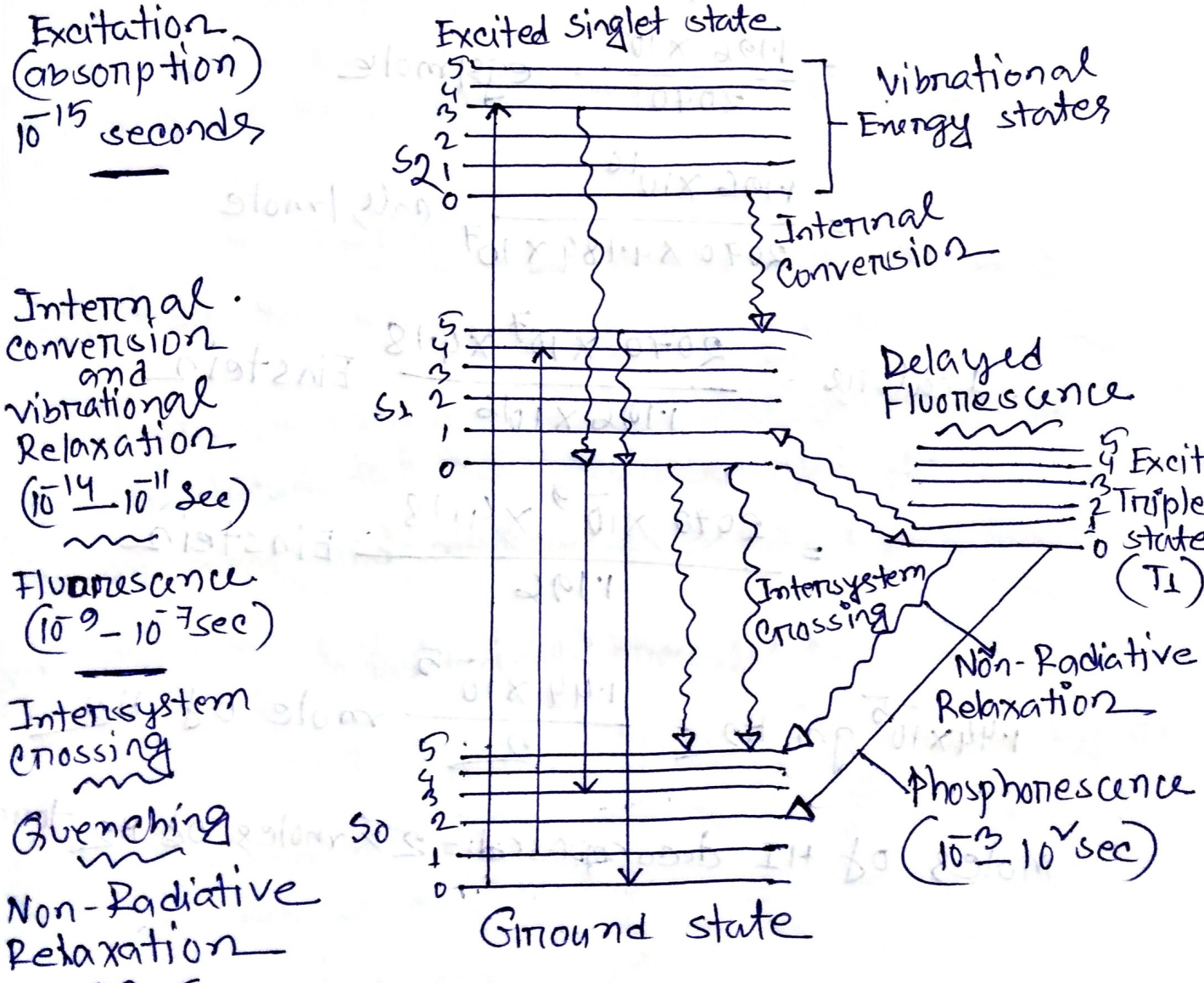
ii) Photolysis of $NH_3$ by a wavelength of 2100 A°

$$2NH_3 + h\nu \longrightarrow N_2 + 3H_2 \; ; \; \phi = 0.2$$

(b) Draw the Jablonski energy diagram. and show the different photo-physical process with their time scale. (4)

Answer

Jablonski Energy Diagram




Scanned with ACE Scanner

(e) The photodecomposition of HI vapour was carried out with $\lambda = 207$ nm radiation. Absorption of each calorie of energy gave $1.44 \times 10^{-5}$ g of hydrogen. What is the quantum yield? ($h = 6.626 \times 10^{-34}$ J s, $c = 3 \times 10^8$ m s$^{-1}$, $N = 6.023 \times 10^{23}$; 1 calorie = 4.184 joules) (2)

$\lambda = 2070\ A^\circ$

**Answer**

$$1 \text{ Einstein} = \frac{1.196 \times 10^{16}}{\lambda} \text{ ergs/mole}$$

$$= \frac{1.196 \times 10^{16}}{2070} \text{ ergs/mole}$$

$$= \frac{1.196 \times 10^{16}}{2070 \times 4.184 \times 10^7} \text{ cals/mole}$$

$$1 \text{ calorie} = \frac{2070 \times 10^7 \times 4.18}{1.196 \times 10^{16}} \text{ Einstein}$$

$$= \frac{2070 \times 10^{-9} \times 4.18}{1.196} \text{ Einstein}$$

$$1.44 \times 10^{-5} \text{ gm } H_2 = \frac{1.44 \times 10^{-5}}{2} \text{ mole of } H_2$$

Moles of HI decomposed = 2 × moles of $H_2$ formed

Scanned with ACE Scanner

$$= \frac{2 \times 1.44 \times 10^{-5}}{2} \text{ moles}$$

$$= 1.44 \times 10^{-5} \text{ moles}$$

$$\phi = \frac{\text{Moles of HI decomposed}}{\text{No. Einstein absorbed}}$$

$$= \frac{1.44 \times 10^{-5} \times 1.96}{2070 \times 4.18 \times 10^{-9}}$$

$$= 1.99 \approx 2$$

(2017)

36 What are the primary and secondary processes? Briefly discuss the consequences of light absorption by an atom and molecule. (3)

Answer Primary photochemical processes :-

Process in which light radiation is absorbed by an atom or molecule giving rise to the formation of an excited atom or excited molecule.

$$A + h\nu \rightarrow A^*$$

Scanned with ACE Scanner

Secondary photochemical processes :-

Process which involves the excited atoms, molecules or free radicals produced in the primary process.

Consequences of light absorption can be discussed under the following two headings:-

① Light absorption by atoms.

② Light absorption by molecules.

① Light absorption by atoms :—

(i) Primary effects of photon absorption by atoms are :-

(a) Electronic excitation.

(b) Ejection of electrons.

(ii) Secondary effects of photon absorption by atoms are :-

(a) Fluorescence.

(b) Resonance fluorescence.

(c) Phosphorescence.

(d) Photochemical reaction.

Scanned with ACE Scanner

(e) Photosensitization.

(2) Consequences of light absorption by molecules:-

(i) Primary effects: The absorption of light may produce the following primary effects:-

(a) Excitation of molecules.

(b) Isomeric change to form new molecules.

(c) Dissociation of molecules.

(d) Ionisation of molecules through electron ejection.

(ii) Secondary effects: If the dissociation or ionisation of absorbing molecules does not occur by direct absorption of radiation, then the energy of excited molecules can give rise to the following secondary effects:

(a) Photosensitization reactions.

(b) Delayed photochemical changes.

(c) Rise in temperature.

(d) Fluorescence.

(e) Phosphorescence.

Scanned with ACE Scanner

(e) The quantum yield of the reaction, $H_{2(g)} + Br_{2(g)} \rightarrow 2HBr_{(g)}$ is $1.0 \times 10^6$ with a wavelength of 480nm. Calculate the number of moles of HBr (g) produced per joule of radiant energy absorbed. (2)

($h = 6.626 \times 10^{-34}$ Js, $c = 3 \times 10^8 ms^{-1}$, $N = 6.022 \times 10^{23}$)

Answer